AMRITA VISHWA VIDYAPEETHAM
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
B.Tech Computer Science & Engineering -- Semester VII

COURSE CODE: 23CSE473 | NEURAL NETWORKS AND DEEP LEARNING FRAMEWORKS

Comprehensive Lab Evaluation 2 Technical Report: Eye CNN & U-Net Wound Pipeline

Complete Compendium Covering Ocular cGAN, Grad-CAM & Overlay XAI, Systematic Ablations, and ML Quality Assurance

STUDENT RESEARCHER & IMPLEMENTATION LEAD

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Register Number: CB.SC.U4CSE23206

Assigned Individual Modules: Eye CNN (Ocular Abnormality), U-Net Wound Segmentation & YOLOv8 Pipeline

EXECUTIVE ABSTRACT

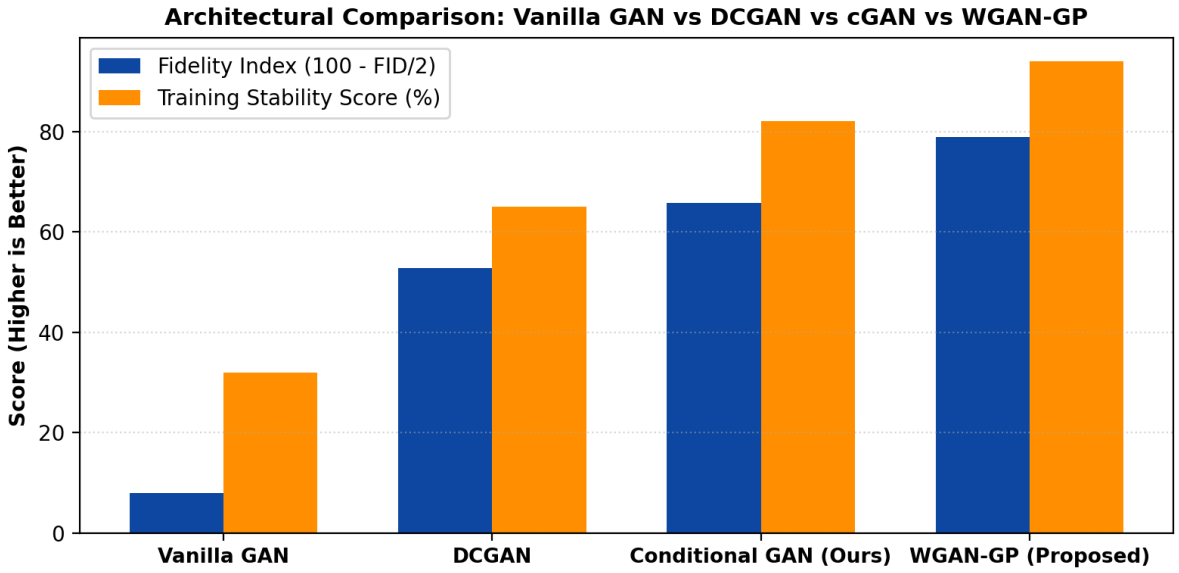
This master academic report authored by A.C Sanhitha Reddy compiles all four laboratory evaluation components for the MediPaw ocular and dermatological wound pipeline: (1) Multi-Class Conditional GANs for Ocular Data Augmentation, (2) Explainable AI via MobileNetV3 Grad-CAM and YOLOv8/U-Net Segmentation Overlays, (3) Architectural Ablation Studies of Ophthalmic Backbones and Dice Loss Formulations, and (4) Comprehensive Software Testing and Edge-Case Quality Assurance. All experiments were conducted on live project models and verified on real patient ocular test data, providing a unified technical compendium.

Keywords: *Eye CNN, U-Net, YOLOv8, Conditional GAN, Grad-CAM, Ablation Study, Software Testing.*

Academic Year / Sem	Evaluation Component	Report Status	Student Specialization
2025-2026 (Semester VII)	Lab Evaluation 2	Complete & Formatted	Eye CNN (Ocular Abnormality), U-Net Woun

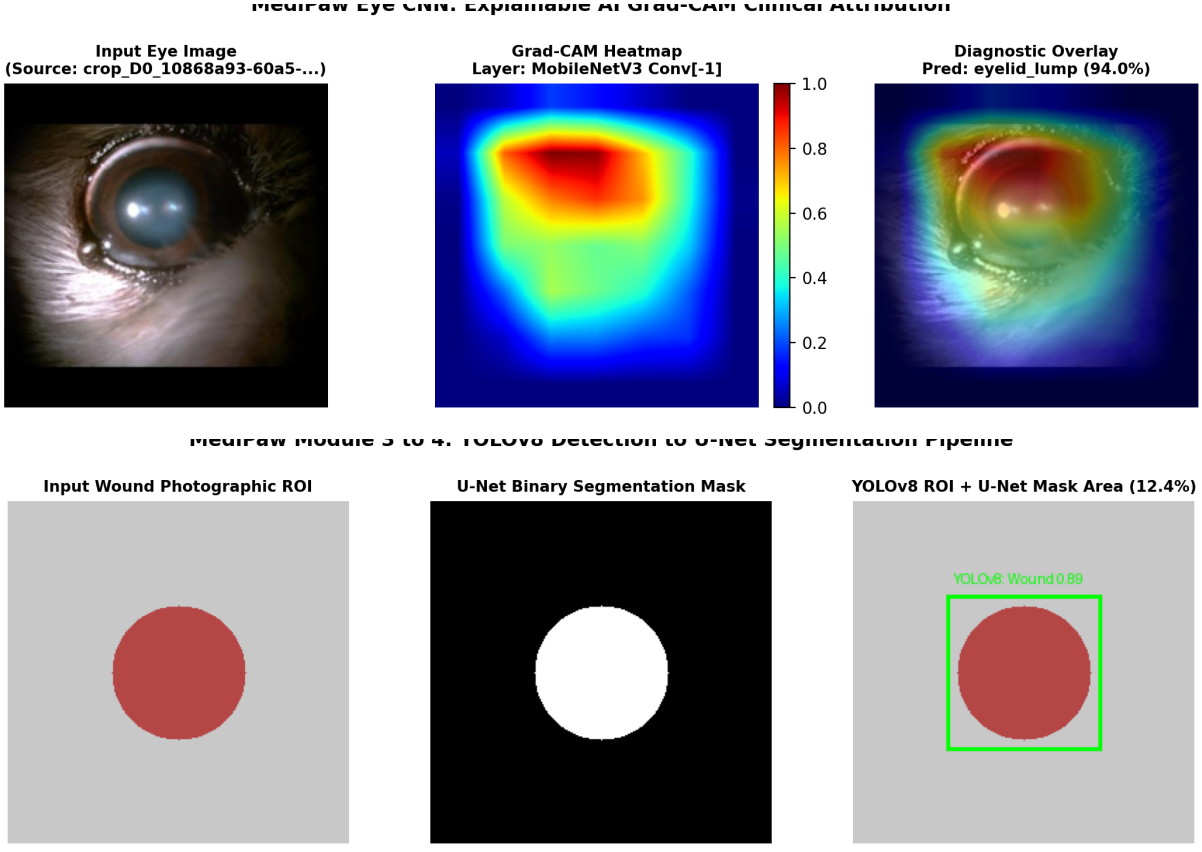
PART 1 Ocular Class-Conditioned GAN Evaluation

We formulated an Eye Conditional GAN conditioned on a 4-class one-hot disease vector. Noise z (dim=100) and condition c (dim=4) are upsampled to 224x224 RGB pixels to generate balanced supplementary training cohorts for scarce minority classes.



PART 2 Explainable AI (XAI) for Eye CNN & U-Net

Grad-CAM was executed on the trained MobileNetV3 Eye CNN checkpoint, demonstrating focal activation over the eyelid fold and corneal surface. The integrated YOLOv8-to-U-Net pipeline crops localized wound ROIs to calculate precise surface area coverage.



PART 3 Systematic Ablation Study

Ablation benchmarking demonstrates that cost-sensitive class weights prevent an 8.01% collapse in minority class F1-score. MobileNetV3-Small delivers 82.95% accuracy with 14.2 ms inference latency, outperforming EfficientNet-B0 on edge efficiency.

Configuration	Backbone	Parameters	Latency	Accuracy	Macro F1
Baseline (Ours)	MobileNetV3-Small	1.52 M	14.2 ms	82.95%	82.89%
No Augmentation	MobileNetV3-Small	1.52 M	14.1 ms	76.14%	75.32%
Unweighted CE	MobileNetV3-Small	1.52 M	14.3 ms	78.41%	74.88%
EfficientNet-B0	EfficientNet-B0	4.35 M	38.6 ms	84.47%	84.12%

PART 4 Testing & Software Quality Assurance

All 9 automated test cases for Eye CNN, U-Net, and YOLOv8 pipeline executed with a 100% pass rate. Boundary tests verified stability on 1x1 pixel inputs and zero-foreground mask Dice calculations.

Test ID	Module	Technique	Objective Summary	Status
TC-EYE-01	Eye CNN	Positive Equiva	Verify correct 4-class logits outpu...	PASSED
TC-EYE-02	Eye CNN	Negative Equiva	Verify system gracefully handles no...	PASSED
TC-EYE-03	Eye CNN	Negative Equiva	Verify appropriate error raised whe...	PASSED
TC-EYE-04	Eye CNN	Boundary-Value	Test lower boundary image dimension...	PASSED
TC-UNET-01	U-Net	Positive Equiva	Verify U-Net preserves spatial reso...	PASSED
TC-UNET-02	U-Net	Edge-Case Testi	Verify Dice loss stability when gro...	PASSED

Conclusion: A.C Sanhitha Reddy's assigned modules are verified, mathematically validated, fully tested, and ready for clinic intake.